

Math 150 - Calculus 1
Homework due 2/18

Directions

Questions 1 and 2 were initially presented in your daily prep assignments for sections 1.2 and 1.3. Question 3 includes activities from the activities workbook that can be found on blackboard [activities workbook that can be found on blackboard](#). Please write up each response neatly and label them clearly. Then use a scanner app on your phone to scan all pages of your work into a single PDF. Make sure each page is clear to read. Then upload your pdf to [gradescope](#). Be sure to select the pages for each entry in the outline.

1. Questions from 1.2.

- (a) Draw an example of a function g so that g does not have a limit at $x = 1$, while g has a limit of -3 at $x = 2$. Write a sentence to explain why your example has the requested properties.
- (b) Consider the function $p(x) = \frac{1-x^2}{1-x}$. What is the most “interesting” x -value at which to wonder if p has a limit? What is the limit of p at that value? Justify your conclusion with appropriate supporting reasoning.

2. Questions from 1.3

- (a) Draw a possible graph of a function g that satisfies $g(2) = 3$ and $g'(2) = -1$. Write one sentence to explain your thinking.
- (b) For $f(x) = x^2$, use the limit definition of the derivative to compute $f'(1)$. Show your work fully.

3. Complete the following activities from the Active Calculus Activities Workbook.

- (a) 1.2.2
- (b) 1.2.3
- (c) 1.3.2